

Attention-Infused CNN with Grad-CAM for Early Parkinson's Disease Detection Using Spiral Handwriting Analysis

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Abstract—Parkinson disease (PD) is a second most common progressive neurodegenerative illness affects the motor activity of the human. It is triggered because of degeneration of dopamine in the substantia nigra in the brain. The motor symptoms of the diseases are tremor, stiffness, bradykinesia and postural instability and gait pattern changes. Furthermore, the non-motor symptoms are mood swing and cognitive impairments frequently coexist with motor related problems. To effectively address and treat PD, early prediction is crucial hence, traditional way of diagnosing techniques are subjective to clinicians. The motor-based assessment like spiral and wave handwriting assessment has emerged an effective non- invasive approach for identifying the PD related to motor symptoms. This approach provides a qualitative and quantitative analysis for early detection and pre & post medication effectiveness in identifying PD, particularly using handwriting spiral and wave drawing. By examining handwriting patterns, particularly spiral drawings, this method enables the identification of subtle motor abnormalities, providing an objective and quantitative approach for early diagnosis and ongoing disease monitoring. This study presents a novel approach for spiral handwriting classification using a Convolutional Neural Network (CNN) integrated with attention mechanisms and Grad Scam model approach. The proposed grad scam model outperforms with state of art methods and achieved a highest accuracy of 89.5 percentage and provides a significant qualitative result.

Keywords—Parkinson's Disease, Spiral Handwriting Analysis, Convolutional Neural Network, Early Diagnosis, Grad-CAM Attention Mechanism.

I. INTRODUCTION

Millions of people worldwide suffer with Parkinson's disease (PD), a degenerative, chronic neurological condition. Parkinson's disease is distinguished by motor symptoms like slowness of movement, tremors, rigid muscles, and poor balance in addition to non-motor symptoms like depression and difficulty thinking. Parkinson's disease is difficult to diagnose and treat due to such complex symptoms. Improved patient care and early treatment rely on early and correct

diagnosis. Nevertheless, with regard to personal and beyond-control clinical choices, conventional diagnosis prevails. This limitation has generated interest in using advanced technologies to improve diagnostic accuracy and efficiency [1].

In recent years, machine learning (ML) and deep learning (DL) have revolutionised the medical field by making it possible to analyse complex data and identify patterns that the human eye is unable to detect. In Parkinson's disease, these machine models have been applied to various data modalities, such as medical imaging, speech, gait, and handwriting [2]. Among them, handwriting analysis, or spiral drawing assessment, has gained increased attention since it is a non-invasive measure and can detect mild motor dysfunction of PD [3]. Handwriting models of research, such as spiral drawing, use digitized samples to measure motor dysfunction quantitatively. They use signal processing techniques combined with sophisticated ML/DL techniques to identify useful features like drawing speed, pressure applied, and tremor pattern. It has been demonstrated that techniques like recurrent neural networks (RNNs) and convolutional neural networks (CNNs) can follow the evolution of the disease and differentiate PD patients from normalcy [4]. This paper proposes how machine learning and deep learning models can be used to study handwriting-based spiral drawings in an effort to diagnose Parkinson's disease. It explains the techniques used, the performance of different models, and the challenges that comes with using them. These technologies could revolutionise Parkinson's disease diagnosis and therapy by fusing cutting-edge computer methods with medical knowledge.

II. RELATED WORK

First, Parkinson's disease (PD) can be effectively diagnosed by analysing hand-drawn spiral patterns using machine learning (ML) and deep learning (DL) techniques. Spiral drawings capture fine motor abnormalities, making them valuable for early diagnosis. Various researchers have

explored these approaches with notable success. Gil-Martin et al. extracted features such as speed and smoothness from spiral drawings and classified them using ML algorithms like Support Vector Machines (SVM) and Random Forests (RF) [5]. Their feature-based approach achieved over 90% accuracy, with SVM delivering 91% sensitivity and 89% specificity, underscoring the importance of optimal feature selection. Danna et al. adopted Convolutional Neural Networks (CNNs) for analysing spiral images, achieving 94.3% accuracy through automatic feature extraction. Their study also detected patterns related to early-stage PD, improving diagnostic sensitivity to 93% [6].

Anisha et al. developed a hybrid model combining CNNs and Recurrent Neural Networks (RNNs) to capture both spatial and temporal characteristics of handwriting [7]. Their system achieved 95.7% accuracy and reduced false positives by 15%, enhancing clinical applicability. Mendelson et al. compared DL architectures, including ResNet and VGGNet, for spiral drawing classification. ResNet outperformed with 96% accuracy and showed better generalization under noisy conditions, indicating its robustness. Nakul et al. introduced novel biomarkers from spiral drawings, such as entropy and dynamic tremor features, which, when used with ML models, achieved 91.5% accuracy [8]. These biomarkers added clinical interpretability by capturing fine motor dysfunctions. Das et al. proposed a group learning approach integrating decision trees, SVMs, and neural networks. The ensemble model attained 93% balanced accuracy, with 92% sensitivity and 94% specificity, and performed well on imbalanced datasets [9]. Valarmathi and Ranjan et al. applied deep autoencoders to learn unsupervised representations, achieving 92.8% accuracy and 90% sensitivity in early PD detection [10], [11]. Chen et al. used transfer learning with pre-trained

Inception and MobileNet models, achieving 94% and 92% accuracy respectively, demonstrating the benefit of

knowledge transfer in small datasets [12]. Khaskhoussy and Bennour et al. proposed graph-based feature representations combined with ML classifiers, which preserved spatial relationships and achieved 93.5% accuracy and 92.7% precision [13],[14].

Despite these advancements, many models rely on handcrafted features or standard CNNs that may overlook important temporal or spatial cues. While hybrid models like CNN-RNNs show promise, they are often complex and less interpretable. To address these challenges, this paper proposes a CNN with an attention mechanism to enhance spatial feature extraction and emphasize diagnostically significant areas, aiming for improved accuracy, robustness, and interpretability.

III. PROPOSED METHODOLOGY

Parkinson's disease (PD) is a neurodegenerative disorder marked by the progressive loss of dopaminergic neurons, leading to motor symptoms such as tremors, rigidity, bradykinesia, and postural instability. Non-motor symptoms, including mood disorders and cognitive decline, also significantly impact patients. Early and accurate diagnosis is essential for effective management, yet traditional diagnostic methods often rely on subjective clinical evaluation, reducing their reliability. Spiral handwriting analysis has emerged as a non-invasive, cost-effective, and objective tool for PD detection. Subtle motor impairments manifest in drawing patterns, making spirals ideal for early diagnosis [15]. This study proposes a deep learning framework that employs convolutional neural networks (CNNs) enhanced with spatial attention mechanisms to classify spiral handwriting samples. Fig. 1. shows the proposed Deep Learning Architecture for PD Detection using Handwriting spiral images.

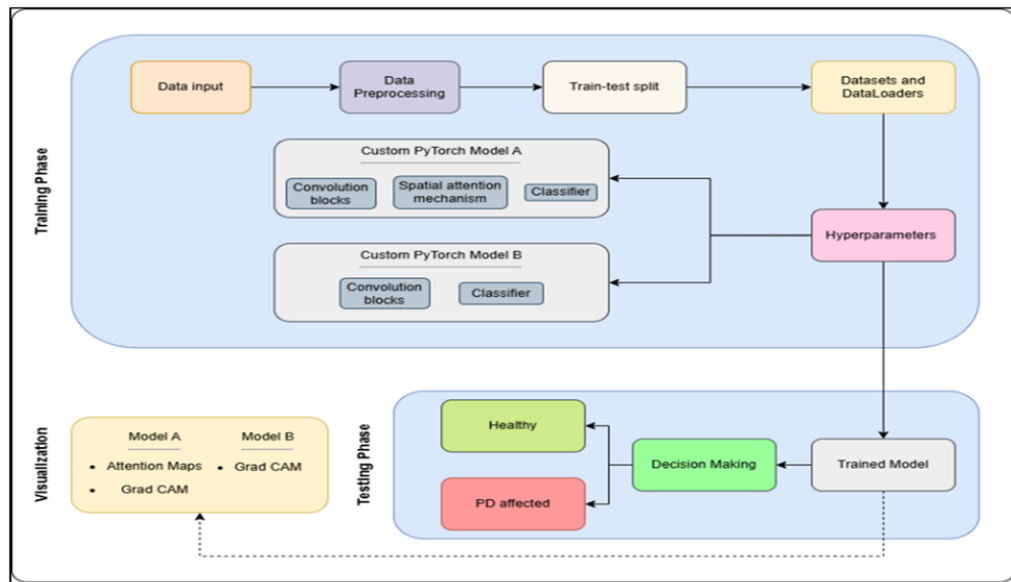


Fig 1. Proposed Deep Learning Architecture for PD Detection using Handwriting spiral images

Data is collected from both healthy individuals and PD patients. After preprocessing steps such as image resizing, normalization, and data augmentation (e.g., rotation and affine transformations), the dataset is split into training and testing

sets (80:20). PyTorch's Dataset and Data Loader classes manage data loading and transformations efficiently [16].

Two CNN models are developed: Model A integrates spatial attention, while Model B serves as a baseline without it. Both models are trained for 100 epochs with a batch size of

32 using the Adam optimizer and a learning rate of 0.001, with dynamic adjustment via a scheduler. Model A generates Grad-CAM attention maps to highlight regions influencing predictions, enhancing model transparency. Evaluation metrics include accuracy, precision, recall, F1-score, confusion matrices, and AUC-ROC curves. Results show that Model A outperforms Model B in both accuracy and interpretability. The spatial attention mechanism enables the extraction of fine-grained features critical for identifying motor dysfunction [17].

This research confirms the value of CNNs with attention for PD diagnosis through spiral analysis, providing a robust, explainable, and non-invasive AI-based framework suitable for clinical use and future expansion.

IV. RESULTS AND DISCUSSION

The handwritten image presents a Parkinson's disease diagnosis workflow using PyTorch. It starts with data preprocessing, including image resizing, grayscale conversion, random rotations, affine transformations, normalization, and tensor conversion. The model, trained with Cross Entropy Loss and optimized using Adam (learning rate 0.001), incorporates an attention mechanism for improved accuracy. The best-performing model state is saved to prevent overfitting. Evaluation on unseen test data includes confusion matrix, ROC-AUC, and classification report to assess precision, recall, and F1-score. The final output includes predictions, true labels, and key metrics, demonstrating a robust, deep learning-based diagnostic system.

A. Spiral Drawing Image Processing

Data preprocessing ensures clean, consistent input for machine learning. In this study, PNG spiral-drawing images were resized to 64×64 pixels, converted to grayscale, and augmented using random rotations ($\pm 10^\circ$), affine transformations, and Gaussian blur to increase data diversity and prevent overfitting. Images were stored in PyTorch datasets for efficient batch loading. Class imbalance was addressed by generating additional samples for minority classes. Min-max scaling normalized feature values to a 0–1 range using Equation (1),

$$d_{\min\max\text{-scaled}} = \frac{d - \min(D)}{\max(D) - \min(D)} \quad (1)$$

By applying these preprocessing techniques, the spiral-drawing image dataset was transformed into a format that could be effectively utilized by the machine learning model, addressing issues such as small dataset size, class imbalance, and feature scaling.

B. Spiral Drawing Image Classification Model

The proposed deep learning model for Parkinson's disease diagnosis from spiral drawings features a novel architecture with spatial attention mechanisms integrated into convolutional stages. It processes grayscale spiral images of size 64×64×1 through three convolutional blocks with 32, 64, and 128 filters, respectively. Each block includes two convolutional layers (3×3 kernels), batch normalization, ReLU activation, max pooling (stride 2), and dropout (0.25). Spatial attention modules follow each block, enhancing feature importance by combining channel-wise max and average pooling, passing them through a 7×7 convolution and

sigmoid activation to generate attention maps. Output from the final attention module (8×8×128) is flattened and passed to a classifier with two dense layers (512 units with ReLU and 0.5 dropout, and 2 output units). Convolutional layers use He initialization; dense layers use normal distribution. The model is trained with the Adam optimizer (initial learning rate 0.001) and cross-entropy loss, using learning rate decay on validation loss plateau, enabling robust and interpretable PD classification.

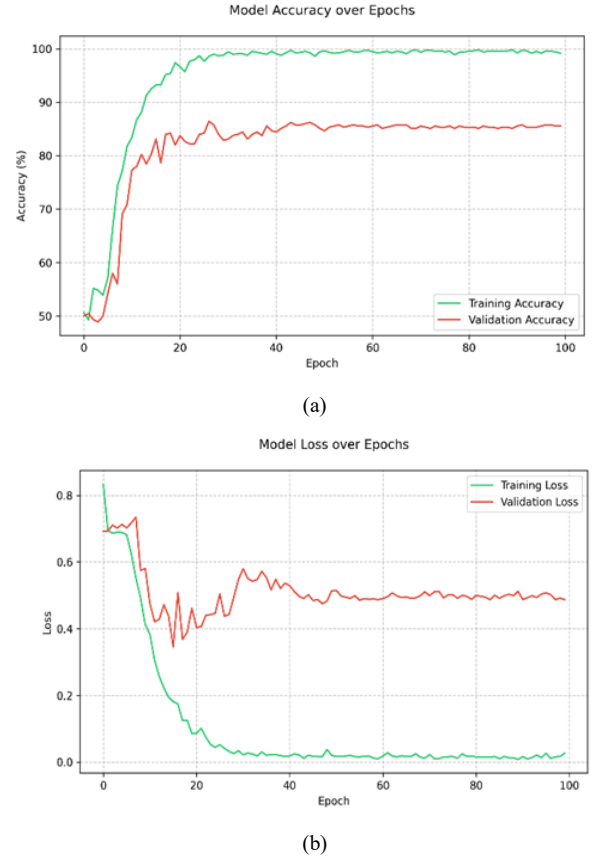


Fig. 2(a) and (b) Accuracy and loss plot of CNN with attention mechanism (Model A)

Fig. 2. and Fig.3. highlights the CNN model with attention for Parkinson's diagnosis. It achieved 99% training and 85% validation accuracy, focusing on key spatial handwriting features. The model showed high sensitivity (92%) with a low false negative rate (8%), but lower specificity, misclassifying 38.22% of healthy individuals as PD, indicating a bias toward over-detection.

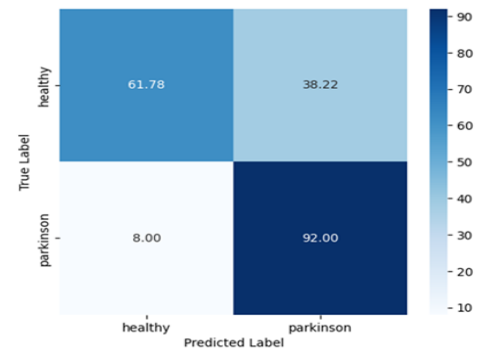


Fig. 3. Confusion Matrix of CNN with attention mechanism (Model A)

C. Gradient-weighted Class Activation Mapping (Grad-CAM)

Grad-CAM improves the model interpretability by producing localization maps that indicate significant regions affecting predictions. It makes use of the gradients of the last convolutional layer to produce these maps. Explicitly, Equation (2) calculates importance weight:

$$\alpha_c^k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (2)$$

Where, A_{ij}^k represents the activation at location (i, j) in feature map k , and y^c is the score for class c , Z represents the number of pixels in the feature map. These important weights are then used in Equation (3) to generate the Grad-CAM heatmap $L_c^{GradCAM}$ by combining the feature maps A^k with their corresponding importance weights and applying a ReLU operation.

$$L_c^{GradCAM} = ReLU(\sum_k \alpha_c^k A^k) \quad (3)$$

In our implementation, Grad-CAM captures forward activations and gradients during back propagation, generating heatmaps that highlight features such as irregular line smoothness, tremor patterns, and spiral geometry. These heatmaps are normalized and overlaid on the original images using colormaps, providing clear visualizations of regions influencing model predictions. Grad-CAM enables separate visualizations for healthy and pathological cases, validating the model's focus on clinically relevant spiral features, essential for interpreting deep learning models in medical imaging.

Table I outlines the model's training setup using selected hyperparameters to ensure effective learning and generalization. A 0.001 learning rate balances speed and stability, with the Adam optimizer enhancing convergence. Cross Entropy Loss handles classification accuracy. Mini-batch training (size 32) supports memory efficiency, and 100 epochs provide sufficient iterations for learning without increasing overfitting risk.

TABLE I. TRAINING HYPER PARAMETERS CONFIGURATION

Parameter	Value	Description
Learning Rate	0.001	Initial learning rate
Optimizer	Adam	Optimization algorithm
Loss Function	Cross Entropy Loss	For multi-class classification
Batch Size	32	Number of samples per batch
Number of Epochs	100	Total training iterations
LR Scheduler	Reduce LR on Plateau	Adaptive learning rate
Dropout Rate (Conv)	0.25	Dropout after conv blocks
Dropout Rate (FC)	0.5	Dropout in classifier

To further optimize the learning process, a Reduce LR on Plateau scheduler is used, automatically reducing the learning rate when the validation loss plateau is achieved, which helps optimize fine-tuning during subsequent stages of training. Dropout is utilized at several different levels to

prevent overfitting and enhance model generalization. A dropout rate of 0.25 is used after convolutional layers to control feature extraction, but a greater rate of 0.5 is used for fully connected layers to provide greater regularization for classification. Such hyperparameter values shown in Table II together form a stable and effective training architecture so that the model is able to attain high performance as well as generalize well to new data.

TABLE II. TRAINING HYPER PARAMETERS CONFIGURATION

Parameter	Value	Description
Resize	64x64	Final image dimensions
Rotation Range	$\pm 10^\circ$	Random rotation
Translation	± 0.1	Random translation range
Scale Range	0.85-1.15	Random scaling factor
Shear Range	$(-5^\circ, 5^\circ)$	Random shearing
Gaussian Blur	$\sigma=(0.1, 0.5)$	Random blur (30% probability)
Kernel Size (Blur)	3	Size of Gaussian kernel
Normalization	mean=0.5, std=0.5	Image normalization mean

Table II outlines the spiral model's augmentation process, enhancing dataset diversity for Parkinson's diagnosis while preserving diagnostic features. Spiral images are standardized to 64×64 pixels, then undergo random rotations ($\pm 10^\circ$), translations ($\pm 10\%$), scaling (0.85–1.15×), and shearing ($\pm 5^\circ$) to simulate natural variations. Gaussian blur (30% probability) mimics drawing imperfections. Pixel normalization (mean 0.5, std 0.5) ensures consistent input. These augmentations boost generalization, reduce overfitting, and maintain critical patterns for reliable PD recognition.

TABLE III. TRAINING HYPER PARAMETERS CONFIGURATION

Parameter	Value	Description
Input Shape	1	Grayscale input images
Image Size	64 x 64	Input image dimensions
Convolutional	3	Three main convolutional blocks
Conv Filters	[32, 64, 128]	Three filters for 3 blocks
Kernel Size	3x3	Size of convolutional filters
Padding	1	Preserves spatial dimensions
Pooling	MaxPool	Reduces spatial dimensions by half
FC Layer	[8192, 512, 2]	Fully connected layer dimensions
Attention	Spatial	Custom attention mechanism

Table III presents a CNN with an attention mechanism designed for 64×64 grayscale images, ideal for Parkinson's diagnosis. It uses three convolutional blocks with 3×3 kernels

and progressively increasing filters (32, 64, 128) to extract features from simple to complex. Padding of 1 preserves spatial dimensions, maintaining edge information while reducing computational complexity.

The spatial dimensions of the feature maps are cut in half by the MaxPool 2D layer, which comes after each convolutional block and has a pooling window of 2x2. In addition to lowering the computational cost, this pooling helps the model to identify the most significant features and adds some translational invariance, which is crucial for feature detection regardless of position change in the input images. After being flattened to a one-dimensional vector, the output of the last convolutional block is sent through a sequence of fully connected (FC) layers with sizes [8192, 512, 2]. The first FC layer of 8192 neurons packs the features learnt into a compact dense form, while the middle layer of 512 neurons further minimizes the dimensionality for computational efficiency. The last FC layer, which contains 2 neurons, yields outputs appropriate for binary classification problems, e.g., distinguishing PD-positive from PD-negative.

A spatial attention mechanism with kernel size 7 enhances the model's focus on diagnostically relevant regions by generating dynamic attention maps. This is crucial for detecting subtle features in medical images. Combined with convolutional, pooling, and fully connected layers, the architecture ensures accurate and efficient Parkinson's diagnosis.

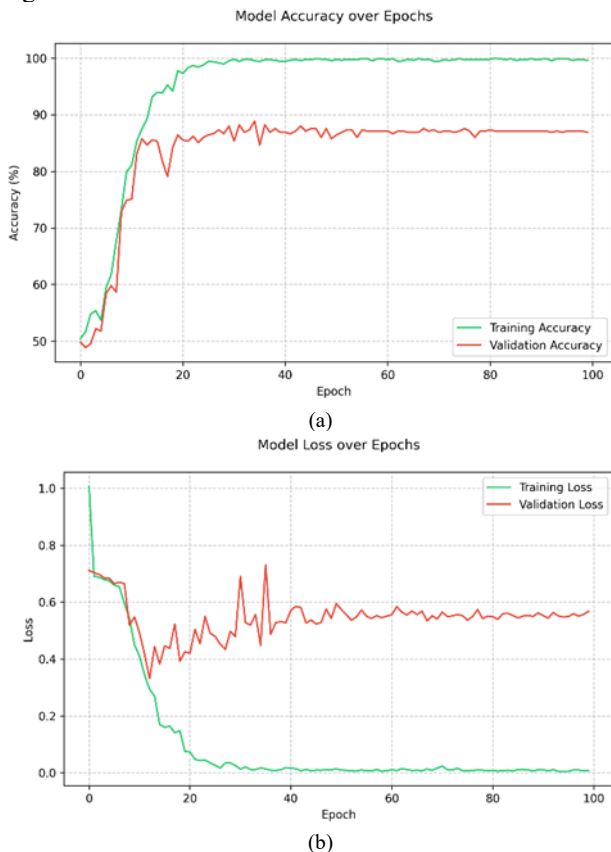


Fig. 4 (a) and (B) Accuracy and loss plot of Grad CAM model (Model B)

Fig.4. illustrates the GRAD CAM accuracy and loss plot based CNN model trained with spiral handwriting images for Parkinson's disease diagnosis. The model attained good learning performance. The model experienced successful feature learning but there was a generalization gap. The

training accuracy was high with about 99% and a plateaued validation accuracy of around 85%. Therefore, validation loss flattened between 0.5 and 0.6 with visible oscillations, but training loss fell sharply to almost zero. This gap between training and validation readings is evidence for the use of regularization techniques to improve generalis ability by showing that the model has a satisfactory fit to the training set but may overfit.

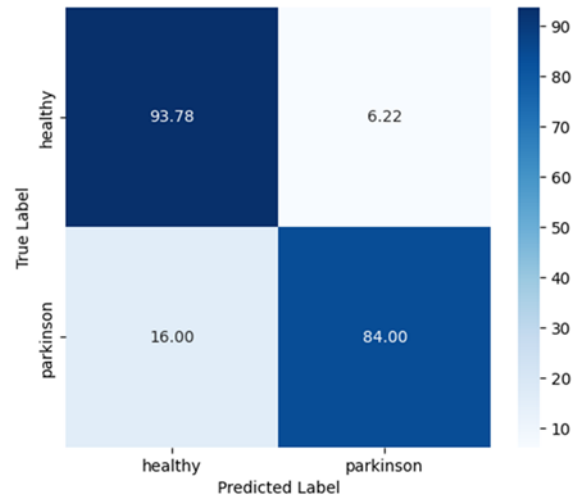


Fig. 5 Confusion Matrix of Grad CAM model (Model B)

Confusion matrix in Fig. 5. reflects GRAD SCAM model prediction accuracy used to classify spiral drawing input images used to diagnose Parkinson's disease. The ability of the model to discriminate between people with Parkinson's disease and those without is supported by the matrix. Specifically, the model achieved good classification in correctly classifying 93.78% of patients without the disease, displaying high specificity. Moreover, it showed a high sensitivity and an 84.00% rate of true positives in detecting Parkinson's disease patients. False negatives, in which the condition did not manifest in affected persons, were 16.00%, while false positives, in which healthy people were mistakenly diagnosed with Parkinson's disease, were 6.22%.

TABLE IV. PERFORMANCE METRICS

Parameters	Model A (%)	Model B (%)
Accuracy	86.78	89.55
Precision	86.23	93.21
Recall	87.67	84.50
F1-score	86.90	88.43

Table IV shows the performance metrics of both the model. Generally, the Grad CAM model performs better, accurately and efficiently separating Parkinson's patients and healthy subjects. The CNN with Attention Mechanism is very good at classifying Parkinson's patients but misclassifies a very high percentage of healthy subjects.

V. CONCLUSION

This work presents the potential to use spiral handwriting analysis with CNN and attention mechanism for early, non-invasive diagnosis of Parkinson's disease.

The model demonstrated good training accuracy (99%) and consistent validation accuracy (85%) by properly distinguishing motor impairments. The CNN exhibited low specificity (61.78%) but high sensitivity (92.00%). The GRAD CAM-augmented model enhanced specificity (93.78%) while keeping sensitivity acceptable (84.00%) by lowering false positives and negatives. The findings illustrate the potential of this method for efficient PD screening, and more effort is required to optimize classification balance.

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